

Urban Sprawl Dynamics Modeling Using CA-MLR-GA Hybrid Approach

5-Year Temporal Resolution | Hyderabad, India | 1995–2050

1. Introduction

Urban sprawl is one of the most significant land-use transformations globally. Understanding and predicting urban expansion patterns is critical for sustainable planning, environmental conservation, and infrastructure development. This proposal outlines a methodology for simulating future urban growth using a hybrid Cellular Automata–Multinomial Logistic Regression–Genetic Algorithm (CA-MLR-GA) approach, leveraging freely available satellite imagery and ancillary datasets processed through Google Earth Engine (GEE). A 5-year temporal resolution (6 epochs, 1995–2020) is adopted to capture non-linear growth dynamics—such as the IT corridor expansion (2000–2005) and ORR construction phase (2005–2010)—that a decadal scheme would average out, while providing four independent validation windows.

2. Research Objectives

1. Generate 6 historical LULC maps (1995–2020, 5-year intervals) using Random Forest classification on Landsat imagery, trained against ESA WorldCover 2020.
2. Compute 5 Markov transition matrices at 5-year resolution to quantify period-specific land-cover conversion rates.
3. Develop a CA-MLR-GA model calibrated on 2015–2020 to simulate spatially explicit transition probabilities.
4. Validate the model across 4 independent temporal windows (1995–2015).
5. Project urban growth for 2025, 2030, and up to 2045/2050 under a business-as-usual scenario.

3. Study Area and Data Sources

3.1 Study Area

The study focuses on a 25 km × 25 km area centred on Hyderabad, India—a rapidly urbanising region that experienced distinct growth phases tied to IT-sector expansion, infrastructure development (ORR, Metro), and policy shifts. All spatial data are resampled to 30m resolution.

3.2 LULC Data

Six Landsat surface reflectance composites (Collection 2, Level-2) are generated using post-monsoon dry-season windows (October–February, ± 1 year) to minimise cloud contamination. Table 1 lists the sensor assignment per epoch.

Table 1: Landsat sensor assignment per epoch.

Epoch	Sensor	Notes
1995	Landsat 5 TM	Clean archive
2000	Landsat 5 + 7	Best coverage
2005	Landsat 5 TM	L7 SLC-off; L5 primary, L7 gap-fill
2010	Landsat 5 + 7	Median composite; L5 degrading
2015	Landsat 8 OLI	Excellent radiometric quality
2020	Landsat 8 OLI	Excellent radiometric quality

A Random Forest classifier is trained on ESA WorldCover 2020 (10m) and applied to all composites, producing four-class maps: Built-up, Vegetation, Agriculture, and Water/Barren. Target accuracy: OA > 85%, Kappa > 0.80.

3.3 Driver Variables

Six driver variables are prepared at the calibration epoch (2015 or 2020) for MLR calibration (Table 2). Terrain drivers (slope, elevation) are time-invariant. Distance to roads is treated as static, consistent with the approach in Vani & Prasad (2021), Khan & Sudheer (2022), and other CA studies, as no multi-temporal road GIS exists for India. Population density and NTL intensity are frozen at the calibration epoch; the Logistic-Trend-CA study (Li et al., 2020) demonstrated that logistic regression suitability surfaces derived from different periods produce nearly identical ROC curves, confirming that terrain and accessibility—the dominant drivers—yield temporally stable spatial relationships.

Table 2: Driver variables (at calibration epoch, 2015 or 2020).

Driver	Source	Resolution	Type
Slope	SRTM DEM	30 m	Static
Elevation	SRTM DEM	30 m	Static
Dist. to Roads	OSM / GRIP v4	30 m	Static
Population Density	GHS-POP R2023A	100 m	Frozen at 2015/2020
NTL Intensity	VIIRS DNB	~500 m	Frozen at 2015/2020
Dist. to CBD	Derived from LULC	30 m	Frozen at 2015/2020

VIF testing is performed to verify multicollinearity (threshold: $VIF > 5$).

4. Methodology: CA-MLR-GA Framework

4.1 LULC Classification

Features: spectral bands (Blue, Green, Red, NIR, SWIR1, SWIR2) and indices (NDVI, NDBI, MNDWI). Random Forest trained on ESA WorldCover 2020; applied identically to all 6 composites. GHS-BUILT-S R2023A (100 m, 5-year epochs) serves as independent cross-validation for the built-up class.

4.2 Markov Chain Demand Estimation

Five 4×4 transition probability matrices are computed from successive LULC pairs (Table 3). Each matrix M_i quantifies the probability that a pixel of class j at t_i transitions to class k at t_{i+1} over a 5-year step.

Table 3: Markov transition matrices and growth context.

Matrix	Period	Growth Context
M_1	1995 $\rightarrow$ 2000	Pre-IT-boom baseline
M_2	2000 $\rightarrow$ 2005	IT corridor expansion; rapid sprawl
M_3	2005 $\rightarrow$ 2010	ORR construction phase
M_4	2010 $\rightarrow$ 2015	Infrastructure-led densification
M_5	2015 $\rightarrow$ 2020	Recent trends

4.3 MLR-Based Transition Probability

A Multinomial Logistic Regression is trained on the **2015 $\rightarrow$ 2020 transition period**, using pixel samples of observed change/persistence and the 6 drivers at the calibration epoch (2015 or 2020):

$$P(\text{class} = k \mid \mathbf{X}) = \frac{\exp(\beta_k \cdot \mathbf{X})}{\sum_j \exp(\beta_j \cdot \mathbf{X})} \quad (1)$$

The output is a **single static transition probability surface** for the built-up class. This surface is reused without recomputation for all validation and projection steps. This is standard practice in CA-LR modelling (Arsanjani et al., 2013; Mirbagheri & Helbich, 2017); the suitability surface captures the stable spatial relationship between drivers and urbanisation, while temporal variation in growth intensity is handled by the period-specific Markov matrices.

4.4 GA-Optimised CA Parameters

A Genetic Algorithm optimises the 3×3 neighbourhood weights (8 parameters) and transition threshold using the **2015 $\rightarrow$ 2020 calibration period**. Starting from observed 2015 LULC, the CA simulates 2020 using the MLR surface and M_5 demand. GA fitness: Kappa + Figure of Merit against observed 2020.

GA parameters: Population = 100, Generations = 500, Crossover = 0.8, Mutation = 0.1, Elite count = 5.

4.5 CA Simulation

The CA iteratively allocates growth based on: (1) the static MLR probability surface, (2) GA-optimised neighbourhood weights, and (3) constraint masks (water bodies). Cells are ranked by combined probability and converted stochastically until the Markov demand quota is satisfied.

4.6 Validation Protocol

The calibration period (2015→2020) is excluded from validation. Four independent windows test the locked model (static MLR surface + fixed CA weights) against different growth regimes by swapping only the Markov matrix:

Table 4: Multi-window validation protocol.

Phase	Start LULC	Matrix	Simulated	vs. Observed
<i>Calibration</i>	2015 obs	M_5	2020 sim	2020 obs (GA target)
Validation 1	1995 obs	M_1	2000 sim	2000 obs
Validation 2	2000 obs	M_2	2005 sim	2005 obs
Validation 3	2005 obs	M_3	2010 sim	2010 obs
Validation 4	2010 obs	M_4	2015 sim	2015 obs

Metrics per window: Overall Accuracy, Kappa, Figure of Merit (changed cells), Producer’s/User’s Accuracy for built-up.

4.7 Projection

Future projection uses observed 2020 LULC as starting state, the static MLR surface, locked CA weights, and M_5 (2015→2020, most recent 5-year rates) as business-as-usual demand. The simulation is iterated in 5-year steps: 2020 $\xrightarrow{M_5}$ 2025 $\xrightarrow{M_5}$ 2030 $\xrightarrow{M_5}$... $\xrightarrow{M_5}$ 2045/2050.

5. Implementation Platform

Data acquisition and preprocessing: Google Earth Engine (GEE). MLR and GA: Python (`scikit-learn`, `DEAP`) via Google Colab. Outputs: GeoTIFF rasters for historical and projected LULC, probability surfaces, and driver importance rankings.

6. Comparative Literature Review

Table 5: Comparison with recent urban growth simulation studies.

Study	Method	Key Drivers	Accuracy	Projection
This study	CA-MLR-GA	Slope, elev., pop., NTL, roads, CBD	Target: K>0.80	2025–2050
Vani & Prasad (2021)	CA-GA / SLEUTH	7 drivers incl. terrain, pop., richness	CA-GA superior	2030, 2050
Khan & Sudheer (2022)	ANN-CA	DEM, slope, roads, pop.	K: 0.93–0.97	2030, 2050
Shafizadeh-M. (2017)	6 ML + CA	8 drivers	AUC: 74.7%	2030
Mhangara (2025)	XGBoost-CA	17 drivers, SHAP	K: 0.93–0.99	2030
Ou et al. (2019)	SDA-CA (DL)	6 drivers	Beat ANN-CA	Validated 2015

7. Expected Outputs

1. 6 historical LULC maps (1995–2020) and projected maps for 2025–2045/2050
2. 5 Markov matrices revealing period-specific transition dynamics
3. Transition probability surface with driver importance ranking
4. 4-window validation report (OA, Kappa, FoM per period)
5. Reproducible GEE scripts and Python notebooks

8. Timeline

9. References

- [1] Arsanjani, J.J. et al. (2013). Integration of logistic regression, Markov chain and cellular automata models to simulate urban expansion. *Int. J. Appl. Earth Obs. Geoinf.*, 21, 265–275.
- [2] Khan, A. & Sudheer, M. (2022). ML-based monitoring and modeling for spatio-temporal urban growth of Islamabad. *Eur. J. Remote Sens.*

Phase	Duration
Data acquisition and preprocessing (GEE)	2 weeks
LULC classification (6 epochs) and accuracy assessment	2 weeks
Driver preparation and VIF testing	1 week
MLR development and GA optimisation	2 weeks
Multi-window validation (4 windows)	2 weeks
Projection and scenario analysis	1 week
Documentation and reporting	2 weeks
Total	12 weeks

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